

# Supplementary Material

## Supplementary Table S1. Trial-level results

Each row represents one trial. T: completion time (s); L: master-side trajectory length (m); O: outcome (S, success; F, failure).

| Block | Operator | Object    | Mode | T (s)   | L (m)  | O |
|-------|----------|-----------|------|---------|--------|---|
| MB03  | P01      | mouse     | A    | 21.9673 | 0.6332 | F |
| MB03  | P01      | mouse     | B    | 17.3734 | 0.6484 | S |
| MB03  | P01      | mouse     | C    | 16.4537 | 0.5699 | S |
| MB03  | P01      | mouse     | D    | 20.7266 | 0.7682 | S |
| MB03  | P01      | mouse     | E    | 22.249  | 0.6481 | S |
| MB09  | P01      | scissors  | A    | 20.659  | 0.5952 | S |
| MB09  | P01      | scissors  | B    | 20.8892 | 0.9875 | F |
| MB09  | P01      | scissors  | D    | 20.8983 | 0.5998 | S |
| MB09  | P01      | scissors  | E    | 23.6364 | 0.7184 | F |
| MB09  | P01      | scissors  | C    | 20.4931 | 0.7578 | S |
| MB06  | P01      | mouse     | A    | 22.1784 | 0.7902 | S |
| MB06  | P01      | mouse     | B    | 19.3855 | 0.6696 | F |
| MB06  | P01      | mouse     | D    | 20.6688 | 0.7218 | S |
| MB06  | P01      | mouse     | E    | 22.0362 | 0.7554 | S |
| MB06  | P01      | mouse     | C    | 19.6104 | 0.6812 | S |
| MB21  | P03      | mouse     | A    | 22.0561 | 0.6905 | S |
| MB21  | P03      | mouse     | B    | 24.5042 | 0.8406 | S |
| MB21  | P03      | mouse     | D    | 20.3404 | 0.8234 | S |
| MB21  | P03      | mouse     | E    | 21.432  | 0.7381 | S |
| MB21  | P03      | mouse     | C    | 22.2288 | 0.7984 | F |
| MB27  | P03      | scissors  | A    | 23.2623 | 0.7705 | S |
| MB27  | P03      | scissors  | B    | 22.077  | 0.8524 | S |
| MB27  | P03      | scissors  | D    | 21.0219 | 0.9843 | F |
| MB27  | P03      | scissors  | E    | 19.568  | 0.6959 | S |
| MB27  | P03      | scissors  | C    | 20.0995 | 0.6642 | S |
| MB24  | P03      | mouse     | A    | 19.7035 | 0.8352 | S |
| MB24  | P03      | mouse     | B    | 24.204  | 1.019  | S |
| MB24  | P03      | mouse     | D    | 21.2149 | 0.6921 | S |
| MB24  | P03      | mouse     | E    | 19.9535 | 0.7165 | S |
| MB24  | P03      | mouse     | C    | 20.8218 | 0.743  | S |
| MB15  | P02      | scissors  | A    | 21.1103 | 0.6776 | F |
| MB15  | P02      | scissors  | B    | 19.7492 | 0.7763 | F |
| MB15  | P02      | scissors  | D    | 20.964  | 0.6468 | S |
| MB15  | P02      | scissors  | E    | 22.8528 | 0.5349 | S |
| MB15  | P02      | scissors  | C    | 17.5832 | 0.7542 | S |
| MB18  | P02      | mouse     | A    | 22.8171 | 0.6597 | S |
| MB18  | P02      | mouse     | B    | 19.1697 | 0.8023 | S |
| MB18  | P02      | mouse     | D    | 20.746  | 0.8665 | S |
| MB18  | P02      | mouse     | E    | 22.5251 | 0.6652 | S |
| MB18  | P02      | mouse     | C    | 19.6294 | 0.8223 | S |
| MB12  | P02      | scissors  | A    | 22.1109 | 0.8229 | S |
| MB12  | P02      | scissors  | B    | 20.1043 | 0.7655 | F |
| MB12  | P02      | scissors  | D    | 24.4447 | 0.6999 | S |
| MB12  | P02      | scissors  | E    | 20.7281 | 0.68   | S |
| MB12  | P02      | scissors  | C    | 17.1904 | 0.6565 | S |
| MB02  | P01      | paper cup | A    | 20.2387 | 0.6952 | S |
| MB02  | P01      | paper cup | B    | 18.834  | 0.6637 | S |
| MB02  | P01      | paper cup | C    | 18.0432 | 0.5838 | S |
| MB02  | P01      | paper cup | D    | 19.1196 | 0.6777 | F |
| MB02  | P01      | paper cup | E    | 19.9499 | 0.6712 | S |
| MB08  | P01      | bottle    | A    | 21.1942 | 0.7547 | S |
| MB08  | P01      | bottle    | B    | 20.6644 | 0.6744 | S |
| MB08  | P01      | bottle    | D    | 20.9106 | 0.7339 | F |
| MB08  | P01      | bottle    | E    | 18.3977 | 0.692  | S |
| MB08  | P01      | bottle    | C    | 19.5792 | 0.6695 | S |
| MB05  | P01      | paper cup | A    | 25.9611 | 0.9493 | S |
| MB05  | P01      | paper cup | B    | 20.1454 | 0.755  | S |
| MB05  | P01      | paper cup | D    | 20.792  | 0.8015 | S |
| MB05  | P01      | paper cup | E    | 21.0563 | 0.794  | S |
| MB05  | P01      | paper cup | C    | 17.8797 | 0.6381 | S |
| MB20  | P03      | paper cup | A    | 20.9114 | 0.9774 | S |
| MB20  | P03      | paper cup | B    | 21.4933 | 0.9173 | F |
| MB20  | P03      | paper cup | D    | 20.8499 | 0.719  | S |
| MB20  | P03      | paper cup | E    | 20.6803 | 0.7488 | F |
| MB20  | P03      | paper cup | C    | 18.5434 | 0.7043 | S |
| MB26  | P03      | bottle    | A    | 24.4515 | 0.9096 | F |
| MB26  | P03      | bottle    | B    | 23.651  | 0.7651 | S |
| MB26  | P03      | bottle    | D    | 21.603  | 0.8028 | S |
| MB26  | P03      | bottle    | E    | 19.9488 | 0.8517 | S |

| Block | Operator | Object    | Mode | T (s)   | L (m)  | O |
|-------|----------|-----------|------|---------|--------|---|
| MB26  | P03      | bottle    | C    | 18.8794 | 0.6972 | S |
| MB23  | P03      | paper cup | A    | 21.5467 | 0.8037 | S |
| MB23  | P03      | paper cup | B    | 19.7902 | 0.7035 | S |
| MB23  | P03      | paper cup | D    | 18.8053 | 0.617  | S |
| MB23  | P03      | paper cup | E    | 23.5113 | 0.6767 | S |
| MB23  | P03      | paper cup | C    | 19.119  | 0.6475 | S |
| MB14  | P02      | bottle    | A    | 21.1828 | 0.7639 | F |
| MB14  | P02      | bottle    | B    | 21.4721 | 0.6454 | S |
| MB14  | P02      | bottle    | D    | 19.9772 | 0.664  | S |
| MB14  | P02      | bottle    | E    | 23.952  | 0.6377 | F |
| MB14  | P02      | bottle    | C    | 21.3193 | 0.7792 | S |
| MB17  | P02      | paper cup | A    | 23.1561 | 0.7838 | F |
| MB17  | P02      | paper cup | B    | 21.0744 | 0.7871 | S |
| MB17  | P02      | paper cup | D    | 22.3201 | 0.8671 | S |
| MB17  | P02      | paper cup | E    | 20.6399 | 0.7132 | S |
| MB17  | P02      | paper cup | C    | 19.8447 | 0.5853 | S |
| MB11  | P02      | bottle    | A    | 21.6166 | 0.6533 | S |
| MB11  | P02      | bottle    | B    | 22.7903 | 0.8687 | S |
| MB11  | P02      | bottle    | D    | 20.6251 | 0.7235 | S |
| MB11  | P02      | bottle    | E    | 18.9052 | 0.6659 | S |
| MB11  | P02      | bottle    | C    | 18.7488 | 0.6005 | S |
| MB01  | P01      | apple     | A    | 19.7647 | 0.6263 | S |
| MB01  | P01      | apple     | B    | 20.3613 | 0.5712 | F |
| MB01  | P01      | apple     | C    | 18.2302 | 0.7079 | S |
| MB01  | P01      | apple     | D    | 20.1232 | 0.7471 | S |
| MB01  | P01      | apple     | E    | 18.8263 | 0.7321 | S |
| MB07  | P01      | banana    | A    | 21.3677 | 0.8979 | S |
| MB07  | P01      | banana    | B    | 20.762  | 0.781  | S |
| MB07  | P01      | banana    | D    | 21.3607 | 0.7123 | S |
| MB07  | P01      | banana    | E    | 18.8917 | 0.7928 | S |
| MB07  | P01      | banana    | C    | 20.6135 | 0.6701 | S |
| MB04  | P01      | banana    | A    | 19.0863 | 0.7552 | S |
| MB04  | P01      | banana    | B    | 20.4131 | 0.835  | S |
| MB04  | P01      | banana    | D    | 20.0671 | 0.6714 | S |
| MB04  | P01      | banana    | E    | 20.3914 | 0.8316 | S |
| MB04  | P01      | banana    | C    | 19.5678 | 0.6649 | S |
| MB19  | P03      | apple     | A    | 20.4341 | 0.7572 | S |
| MB19  | P03      | apple     | B    | 21.9394 | 0.852  | S |
| MB19  | P03      | apple     | D    | 21.7007 | 0.7223 | S |
| MB19  | P03      | apple     | E    | 22.138  | 0.7477 | S |
| MB19  | P03      | apple     | C    | 20.4061 | 0.6901 | S |
| MB25  | P03      | banana    | A    | 18.4814 | 0.7055 | S |
| MB25  | P03      | banana    | B    | 21.0957 | 0.907  | S |
| MB25  | P03      | banana    | D    | 20.7293 | 0.6453 | S |
| MB25  | P03      | banana    | E    | 20.1558 | 0.8482 | S |
| MB25  | P03      | banana    | C    | 18.4779 | 0.8467 | S |
| MB22  | P03      | banana    | A    | 20.584  | 0.8097 | S |
| MB22  | P03      | banana    | B    | 21.7273 | 0.9022 | S |
| MB22  | P03      | banana    | D    | 20.2005 | 0.7632 | S |
| MB22  | P03      | banana    | E    | 21.2047 | 0.8606 | S |
| MB22  | P03      | banana    | C    | 19.6026 | 0.7299 | S |
| MB13  | P02      | apple     | A    | 20.6792 | 0.717  | S |
| MB13  | P02      | apple     | B    | 20.8904 | 0.8292 | S |
| MB13  | P02      | apple     | D    | 21.1041 | 0.7061 | S |
| MB13  | P02      | apple     | E    | 20.5426 | 0.804  | S |
| MB13  | P02      | apple     | C    | 18.3496 | 0.8878 | S |
| MB16  | P02      | banana    | A    | 20.8514 | 0.8434 | S |
| MB16  | P02      | banana    | B    | 20.2425 | 1.0097 | S |
| MB16  | P02      | banana    | D    | 22.9636 | 0.7686 | S |
| MB16  | P02      | banana    | E    | 22.5321 | 0.9161 | S |
| MB16  | P02      | banana    | C    | 19.1901 | 0.9025 | S |
| MB10  | P02      | apple     | A    | 20.954  | 0.7275 | S |
| MB10  | P02      | apple     | B    | 22.4177 | 0.7386 | S |
| MB10  | P02      | apple     | D    | 20.3035 | 0.678  | S |
| MB10  | P02      | apple     | E    | 22.2594 | 0.8061 | S |
| MB10  | P02      | apple     | C    | 19.9973 | 0.8459 | S |

**Supplementary Table S2. Preassigned mode order**

| Operator/group | Chronological mode order                                    |
|----------------|-------------------------------------------------------------|
| P01-group 1    | $A \rightarrow D \rightarrow C \rightarrow B \rightarrow E$ |
| P01-group 2    | $B \rightarrow E \rightarrow D \rightarrow C \rightarrow A$ |
| P01-group 3    | $C \rightarrow A \rightarrow E \rightarrow D \rightarrow B$ |
| P02-group 4    | $D \rightarrow B \rightarrow C \rightarrow E \rightarrow A$ |
| P02-group 5    | $E \rightarrow C \rightarrow A \rightarrow B \rightarrow D$ |
| P02-group 6    | $A \rightarrow D \rightarrow B \rightarrow C \rightarrow E$ |
| P03-group 7    | $B \rightarrow E \rightarrow A \rightarrow D \rightarrow C$ |
| P03-group 8    | $C \rightarrow B \rightarrow D \rightarrow A \rightarrow E$ |
| P03-group 9    | $D \rightarrow A \rightarrow E \rightarrow C \rightarrow B$ |